

HAL CANON

BOOK V

Operations Manual

Version 1.0 • Certified final

HAL Book V — Operations Manual

Version: 1.0

Status: CERTIFIED FINAL

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Authority Statement

Book I is supreme. Books II-IV define architecture, engineering, and components. Book IX defines contracts. Book VIII governs verification and certification. Book VI governs the continuing security, privacy, and trust program. Book X controls terminology. Book V defines repeatable operations and creates none of those authorities.

Revision History

Contents

1. Authority, Environments, Roles, and Operational Governance
2. Installation, Bootstrap, Startup, Shutdown, and Runtime Modes
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11. Maintenance, Upgrade, Patch, Deprecation, and Removal
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Conformance model

A procedure conforms only when its preconditions, ordered actions, stop conditions, evidence, recovery, verification, and approvals pass. Critical control failure blocks the action or continued reliance. Higher-order requirements cannot be waived.

Chapter 1 — Authority, Environments, Roles, and Operational Governance

Status: FINAL

Responsible role: Operations Manager

Owner Review items: None

Purpose

Establish controlled operational authority, environment classes, role separation, handoffs, evidence, and stop-work rules.

Scope

This chapter applies across development, test, staging, recovery, degraded environments, and every declared live-effect environment and approved Reality Boundary stage when the described operational condition exists.

Authority and prerequisites

Books I-IV, IX, and X govern. Book VIII controls verification and certification. Book VI controls the continuing security, privacy, and trust program after reconciliation. Required prerequisites are authenticated operator identity, explicit Authority, exact target identification, an approved change or incident record, current certification, evidence capture, and a tested containment or recovery path.

Normative controls

OPS-GOV-01-01 — Every operational action

Every operational action **MUST** identify the accountable role, acting identity, authority source, scope, environment, and evidence record.

Applicability: Authority, Environments, Roles, and Operational Governance. Responsible: Operations Manager.

Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.

Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-GOV-01-02 — Duties in the declared live-effect environment and approved Reality Boun

Duties in the declared live-effect environment and approved Reality Boundary stage **MUST** separate request, approval, execution, and verification for protected or high-risk changes.

Applicability: Authority, Environments, Roles, and Operational Governance. Responsible: Operations Manager. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-GOV-01-03 — Operators

Operators **MUST** stop when identity, authority, certification, Treaty, integrity, target, or real-world state is uncertain.

Applicability: Authority, Environments, Roles, and Operational Governance. Responsible: Operations Manager. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-GOV-01-04 — Operational exceptions

Operational exceptions **MUST** be explicit, risk-assessed, compensating, approved, expiring, reviewable, and automatically invalid after expiry.

Applicability: Authority, Environments, Roles, and Operational Governance. Responsible: Operations Manager. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

Mandatory procedure

1. Pre-authorize. Confirm ticket, risk class, acting identity, authority, separation of duties, current certification, approved window, target, and rollback/containment.
2. Baseline. Capture health, state versions, queues, capacity, active transactions, Treaties, alerts, evidence integrity, and human-visible conditions.
3. Validate inputs. Verify signed artifacts/configuration, Book IX compatibility, dependencies, credentials by reference, privacy classification, and expected state.
4. Establish safety. Enable observation, freeze conflicting work, bound blast radius, assign commander and verifier, and announce stop conditions.

5. Execute. Perform the approved action one checkpoint at a time; record operator, timestamp, command or contract, target, result, and evidence digest.
6. Observe. Compare health, outcomes, invariants, authority decisions, privacy signals, queue behavior, and resource use to approved thresholds.
7. Decide. Continue only when success criteria pass and no stop condition fires; otherwise halt, contain, roll back, or enter forward recovery.
8. Reconcile. Resolve state, messages, external effects, in-flight transactions, evidence gaps, and temporary access; do not guess ambiguous reality.
9. Verify. Run targeted Book VIII verification, contract tests, recovery checks, and independent review appropriate to risk.
10. Close. Record final state, limitations, certificate impact, follow-up actions, approvers, evidence manifest, and communication; revoke temporary authority.

Stop conditions

Stop immediately on identity or authority uncertainty, certificate suspension/revocation/expiry, source or target mismatch, integrity failure, unapproved Treaty or Firewall rejection, unexpected protected-state mutation, unbounded queue/resource growth, privacy exposure, ambiguous external effect, failed invariant, or loss of evidence capture.

Rollback, forward recovery, and escalation

Use rollback only when its preconditions are proven and it cannot repeat or hide a committed Reality Boundary effect. Otherwise contain, preserve evidence, reconcile actual state, and use the approved forward-recovery path. Escalate Critical conditions immediately to the Incident Commander, Security Incident Commander, Certification Authority, and constitutional steward as applicable.

Evidence and completion criteria

Completion requires reconciled authoritative state and external effects, satisfied health and outcome gates, closed temporary access, current certification disposition, retained evidence manifest, independent verifier approval, communication to affected Principals, humans, and stakeholders when material, and recorded follow-up ownership.

Examples and anti-patterns

Example: the operator halts a rollout when authority-decision latency rises and evidence gaps appear, preserves the canary, and requests targeted reverification. Anti-pattern: declaring success because processes are live while certification is suspended or state remains unreconciled.

Traceability and review

Constitutional, architecture, engineering, interface, verification, semantic, security/privacy/trust, reliability, and practicability reviews: PASS after final Books VI, VIII, and IX reconciliation. This chapter does not redefine components, contracts, certification authority, or canonical meaning.

Chapter 2 — Installation, Bootstrap, Startup, Shutdown, and Runtime Modes

Status: FINAL

Responsible role: Runtime Operator

Owner Review items: None

Purpose

Operate installation and lifecycle transitions without bypassing identity, authority, certification, or constitutional admission.

Scope

This chapter applies across development, test, staging, recovery, degraded environments, and every declared live-effect environment and approved Reality Boundary stage when the described operational condition exists.

Authority and prerequisites

Books I-IV, IX, and X govern. Book VIII controls verification and certification. Book VI controls the continuing security, privacy, and trust program after reconciliation. Required prerequisites are authenticated operator identity, explicit Authority, exact target identification, an approved change or incident record, current certification, evidence capture, and a tested containment or recovery path.

Normative controls

OPS-LIF-02-01 — Installation

Installation MUST verify artifact signatures, provenance, compatibility, approved topology, secrets references, and environment classification before execution.

Applicability: Installation, Bootstrap, Startup, Shutdown, and Runtime Modes. Responsible: Runtime Operator.

Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.

Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-LIF-02-02 — Bootstrap

Bootstrap **MUST** establish Constitutional Kernel, identity, authority, time, evidence, policy, and trust prerequisites before admitting ordinary workloads.

Applicability: Installation, Bootstrap, Startup, Shutdown, and Runtime Modes. Responsible: Runtime Operator.

Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.

Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-LIF-02-03 — Startup

Startup **MUST** verify required dependencies, state integrity, contract compatibility, certification status, and an authorized runtime mode with declared entry criteria before traffic admission.

Applicability: Installation, Bootstrap, Startup, Shutdown, and Runtime Modes. Responsible: Runtime Operator.

Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.

Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-LIF-02-04 — Shutdown

Shutdown **MUST** quiesce new work, reconcile in-flight effects, persist required evidence, preserve identity continuity, and verify bounded termination.

Applicability: Installation, Bootstrap, Startup, Shutdown, and Runtime Modes. Responsible: Runtime Operator.

Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.

Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

Mandatory procedure

1. Pre-authorize. Confirm ticket, risk class, acting identity, authority, separation of duties, current certification, approved window, target, and rollback/containment.
2. Baseline. Capture health, state versions, queues, capacity, active transactions, Treaties, alerts, evidence integrity, and human-visible conditions.
3. Validate inputs. Verify signed artifacts/configuration, Book IX compatibility, dependencies, credentials by reference, privacy classification, and expected state.
4. Establish safety. Enable observation, freeze conflicting work, bound blast radius, assign commander and verifier, and announce stop conditions.
5. Execute. Perform the approved action one checkpoint at a time; record operator, timestamp, command or contract, target, result, and evidence digest.

6. Observe. Compare health, outcomes, invariants, authority decisions, privacy signals, queue behavior, and resource use to approved thresholds.
7. Decide. Continue only when success criteria pass and no stop condition fires; otherwise halt, contain, roll back, or enter forward recovery.
8. Reconcile. Resolve state, messages, external effects, in-flight transactions, evidence gaps, and temporary access; do not guess ambiguous reality.
9. Verify. Run targeted Book VIII verification, contract tests, recovery checks, and independent review appropriate to risk.
10. Close. Record final state, limitations, certificate impact, follow-up actions, approvers, evidence manifest, and communication; revoke temporary authority.

Stop conditions

Stop immediately on identity or authority uncertainty, certificate suspension/revocation/expiry, source or target mismatch, integrity failure, unapproved Treaty or Firewall rejection, unexpected protected-state mutation, unbounded queue/resource growth, privacy exposure, ambiguous external effect, failed invariant, or loss of evidence capture.

Rollback, forward recovery, and escalation

Use rollback only when its preconditions are proven and it cannot repeat or hide a committed Reality Boundary effect. Otherwise contain, preserve evidence, reconcile actual state, and use the approved forward-recovery path. Escalate Critical conditions immediately to the Incident Commander, Security Incident Commander, Certification Authority, and constitutional steward as applicable.

Evidence and completion criteria

Completion requires reconciled authoritative state and external effects, satisfied health and outcome gates, closed temporary access, current certification disposition, retained evidence manifest, independent verifier approval, communication to affected Principals, humans, and stakeholders when material, and recorded follow-up ownership.

Examples and anti-patterns

Example: the operator halts a rollout when authority-decision latency rises and evidence gaps appear, preserves the canary, and requests targeted reverification. Anti-pattern: declaring success because processes are live while certification is suspended or state remains unreconciled.

Traceability and review

Constitutional, architecture, engineering, interface, verification, semantic, security/privacy/trust, reliability, and practicability reviews: PASS after final Books VI, VIII, and IX reconciliation. This chapter does not redefine components, contracts, certification authority, or canonical meaning.

Chapter 3 — Configuration, Credentials, Secrets, Certificates, and Keys

Status: FINAL

Responsible role: Platform Operator

Owner Review items: None

Purpose

Control configuration and protected material through authorized, attributable, reversible, and tested procedures.

Scope

This chapter applies across development, test, staging, recovery, degraded environments, and every declared live-effect environment and approved Reality Boundary stage when the described operational condition exists.

Authority and prerequisites

Books I-IV, IX, and X govern. Book VIII controls verification and certification. Book VI controls the continuing security, privacy, and trust program after reconciliation. Required prerequisites are authenticated operator identity, explicit Authority, exact target identification, an approved change or incident record, current certification, evidence capture, and a tested containment or recovery path.

Normative controls

OPS-CFG-03-01 — Configuration

Configuration **MUST** be versioned, reviewed, signed where required, environment-scoped, schema-valid, and attributable.

Applicability: Configuration, Credentials, Secrets, Certificates, and Keys. Responsible: Platform Operator.

Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.

Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-CFG-03-02 — Secrets

Secrets **MUST** never appear in source, logs, tickets, chat, command history, images, metrics, or unrestricted evidence.

Applicability: Configuration, Credentials, Secrets, Certificates, and Keys. Responsible: Platform Operator.
 Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.
 Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-CFG-03-03 — Key and certificate operations

Key and certificate operations **MUST** use dual control, approved algorithms, protected custody, rotation overlap, revocation, and verification.

Applicability: Configuration, Credentials, Secrets, Certificates, and Keys. Responsible: Platform Operator.
 Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.
 Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-CFG-03-04 — Configuration drift

Configuration drift **MUST** trigger detection, classification, reconciliation, and certification-impact review before normalization.

Applicability: Configuration, Credentials, Secrets, Certificates, and Keys. Responsible: Platform Operator.
 Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.
 Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

Mandatory procedure

1. Pre-authorize. Confirm ticket, risk class, acting identity, authority, separation of duties, current certification, approved window, target, and rollback/containment.
2. Baseline. Capture health, state versions, queues, capacity, active transactions, Treaties, alerts, evidence integrity, and human-visible conditions.
3. Validate inputs. Verify signed artifacts/configuration, Book IX compatibility, dependencies, credentials by reference, privacy classification, and expected state.
4. Establish safety. Enable observation, freeze conflicting work, bound blast radius, assign commander and verifier, and announce stop conditions.
5. Execute. Perform the approved action one checkpoint at a time; record operator, timestamp, command or contract, target, result, and evidence digest.
6. Observe. Compare health, outcomes, invariants, authority decisions, privacy signals, queue behavior, and resource use to approved thresholds.
7. Decide. Continue only when success criteria pass and no stop condition fires; otherwise halt, contain, roll back, or enter forward recovery.

8. Reconcile. Resolve state, messages, external effects, in-flight transactions, evidence gaps, and temporary access; do not guess ambiguous reality.
9. Verify. Run targeted Book VIII verification, contract tests, recovery checks, and independent review appropriate to risk.
10. Close. Record final state, limitations, certificate impact, follow-up actions, approvers, evidence manifest, and communication; revoke temporary authority.

Stop conditions

Stop immediately on identity or authority uncertainty, certificate suspension/revocation/expiry, source or target mismatch, integrity failure, unapproved Treaty or Firewall rejection, unexpected protected-state mutation, unbounded queue/resource growth, privacy exposure, ambiguous external effect, failed invariant, or loss of evidence capture.

Rollback, forward recovery, and escalation

Use rollback only when its preconditions are proven and it cannot repeat or hide a committed Reality Boundary effect. Otherwise contain, preserve evidence, reconcile actual state, and use the approved forward-recovery path. Escalate Critical conditions immediately to the Incident Commander, Security Incident Commander, Certification Authority, and constitutional steward as applicable.

Evidence and completion criteria

Completion requires reconciled authoritative state and external effects, satisfied health and outcome gates, closed temporary access, current certification disposition, retained evidence manifest, independent verifier approval, communication to affected Principals, humans, and stakeholders when material, and recorded follow-up ownership.

Examples and anti-patterns

Example: the operator halts a rollout when authority-decision latency rises and evidence gaps appear, preserves the canary, and requests targeted reverification. Anti-pattern: declaring success because processes are live while certification is suspended or state remains unreconciled.

Traceability and review

Constitutional, architecture, engineering, interface, verification, semantic, security/privacy/trust, reliability, and practicability reviews: PASS after final Books VI, VIII, and IX reconciliation. This chapter does not redefine components, contracts, certification authority, or canonical meaning.

Chapter 4 — Deployment, Release Admission, Change Windows, and Rollout

Status: FINAL

Responsible role: Release Manager

Owner Review items: None

Purpose

Admit only signed, compatible, certified artifacts and execute bounded progressive delivery.

Scope

This chapter applies across development, test, staging, recovery, degraded environments, and every declared live-effect environment and approved Reality Boundary stage when the described operational condition exists.

Authority and prerequisites

Books I-IV, IX, and X govern. Book VIII controls verification and certification. Book VI controls the continuing security, privacy, and trust program after reconciliation. Required prerequisites are authenticated operator identity, explicit Authority, exact target identification, an approved change or incident record, current certification, evidence capture, and a tested containment or recovery path.

Normative controls

OPS-DEP-04-01 — A release

A release **MUST** be signed, provenance-verifiable, vulnerability-disposed, contract-compatible, migration-ready, rollback-capable, and currently certified.

Applicability: Deployment, Release Admission, Change Windows, and Rollout. Responsible: Release Manager.

Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.

Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-DEP-04-02 — Deployment

Deployment **MUST** use a declared change window, risk class, blast-radius bound, observation gates, stop conditions, and accountable commander.

Applicability: Deployment, Release Admission, Change Windows, and Rollout. Responsible: Release Manager. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-DEP-04-03 — Traffic progression

Traffic progression **MUST** follow the approved verification rung and **MUST** halt or regress when success, harm, authority, evidence, or health criteria fail.

Applicability: Deployment, Release Admission, Change Windows, and Rollout. Responsible: Release Manager. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-DEP-04-04 — Deployment success

Deployment success **MUST** require post-release validation of outcomes, invariants, authority paths, trust boundaries, privacy duties, and recovery readiness.

Applicability: Deployment, Release Admission, Change Windows, and Rollout. Responsible: Release Manager. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

Mandatory procedure

1. Pre-authorize. Confirm ticket, risk class, acting identity, authority, separation of duties, current certification, approved window, target, and rollback/containment.
2. Baseline. Capture health, state versions, queues, capacity, active transactions, Treaties, alerts, evidence integrity, and human-visible conditions.
3. Validate inputs. Verify signed artifacts/configuration, Book IX compatibility, dependencies, credentials by reference, privacy classification, and expected state.
4. Establish safety. Enable observation, freeze conflicting work, bound blast radius, assign commander and verifier, and announce stop conditions.
5. Execute. Perform the approved action one checkpoint at a time; record operator, timestamp, command or contract, target, result, and evidence digest.
6. Observe. Compare health, outcomes, invariants, authority decisions, privacy signals, queue behavior, and resource use to approved thresholds.
7. Decide. Continue only when success criteria pass and no stop condition fires; otherwise halt, contain, roll back, or enter forward recovery.

8. Reconcile. Resolve state, messages, external effects, in-flight transactions, evidence gaps, and temporary access; do not guess ambiguous reality.
9. Verify. Run targeted Book VIII verification, contract tests, recovery checks, and independent review appropriate to risk.
10. Close. Record final state, limitations, certificate impact, follow-up actions, approvers, evidence manifest, and communication; revoke temporary authority.

Stop conditions

Stop immediately on identity or authority uncertainty, certificate suspension/revocation/expiry, source or target mismatch, integrity failure, unapproved Treaty or Firewall rejection, unexpected protected-state mutation, unbounded queue/resource growth, privacy exposure, ambiguous external effect, failed invariant, or loss of evidence capture.

Rollback, forward recovery, and escalation

Use rollback only when its preconditions are proven and it cannot repeat or hide a committed Reality Boundary effect. Otherwise contain, preserve evidence, reconcile actual state, and use the approved forward-recovery path. Escalate Critical conditions immediately to the Incident Commander, Security Incident Commander, Certification Authority, and constitutional steward as applicable.

Evidence and completion criteria

Completion requires reconciled authoritative state and external effects, satisfied health and outcome gates, closed temporary access, current certification disposition, retained evidence manifest, independent verifier approval, communication to affected Principals, humans, and stakeholders when material, and recorded follow-up ownership.

Examples and anti-patterns

Example: the operator halts a rollout when authority-decision latency rises and evidence gaps appear, preserves the canary, and requests targeted reverification. Anti-pattern: declaring success because processes are live while certification is suspended or state remains unreconciled.

Traceability and review

Constitutional, architecture, engineering, interface, verification, semantic, security/privacy/trust, reliability, and practicability reviews: PASS after final Books VI, VIII, and IX reconciliation. This chapter does not redefine components, contracts, certification authority, or canonical meaning.

Chapter 5 — Cluster, Workload, Messaging, State, and Persistence Operations

Status: FINAL

Responsible role: Platform Operator

Owner Review items: None

Purpose

Operate distributed placement, messaging, state, and persistence while preserving Book IV ownership and Book IX contracts.

Scope

This chapter applies across development, test, staging, recovery, degraded environments, and every declared live-effect environment and approved Reality Boundary stage when the described operational condition exists.

Authority and prerequisites

Books I-IV, IX, and X govern. Book VIII controls verification and certification. Book VI controls the continuing security, privacy, and trust program after reconciliation. Required prerequisites are authenticated operator identity, explicit Authority, exact target identification, an approved change or incident record, current certification, evidence capture, and a tested containment or recovery path.

Normative controls

OPS-CLU-05-01 — Placement

Placement **MUST** respect identity, trust zone, capability, data classification, resource, locality, resilience, and certification constraints.

Applicability: Cluster, Workload, Messaging, State, and Persistence Operations. Responsible: Platform Operator. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-CLU-05-02 — Operators

Operators **MUST NOT** bypass the Event and Messaging Platform or State and Persistence Platform to mutate another component's authoritative state.

Applicability: Cluster, Workload, Messaging, State, and Persistence Operations. Responsible: Platform Operator. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-CLU-05-03 — Messaging operations

Messaging operations **MUST** preserve Book IX contract, ordering, idempotency, deadline, retry, provenance, and dead-letter semantics.

Applicability: Cluster, Workload, Messaging, State, and Persistence Operations. Responsible: Platform Operator. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-CLU-05-04 — Cluster repair

Cluster repair **MUST** prevent split-brain identity, authority, state ownership, duplicate Reality Boundary effects, and unverified member admission.

Applicability: Cluster, Workload, Messaging, State, and Persistence Operations. Responsible: Platform Operator. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

Mandatory procedure

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4. Establish safety. Enable observation, freeze conflicting work, bound blast radius, assign commander and verifier, and announce stop conditions.
5. Execute. Perform the approved action one checkpoint at a time; record operator, timestamp, command or contract, target, result, and evidence digest.
6. Observe. Compare health, outcomes, invariants, authority decisions, privacy signals, queue behavior, and resource use to approved thresholds.
7. Decide. Continue only when success criteria pass and no stop condition fires; otherwise halt, contain, roll back, or enter forward recovery.

8. Reconcile. Resolve state, messages, external effects, in-flight transactions, evidence gaps, and temporary access; do not guess ambiguous reality.
9. Verify. Run targeted Book VIII verification, contract tests, recovery checks, and independent review appropriate to risk.
10. Close. Record final state, limitations, certificate impact, follow-up actions, approvers, evidence manifest, and communication; revoke temporary authority.

Stop conditions

Stop immediately on identity or authority uncertainty, certificate suspension/revocation/expiry, source or target mismatch, integrity failure, unapproved Treaty or Firewall rejection, unexpected protected-state mutation, unbounded queue/resource growth, privacy exposure, ambiguous external effect, failed invariant, or loss of evidence capture.

Rollback, forward recovery, and escalation

Use rollback only when its preconditions are proven and it cannot repeat or hide a committed Reality Boundary effect. Otherwise contain, preserve evidence, reconcile actual state, and use the approved forward-recovery path. Escalate Critical conditions immediately to the Incident Commander, Security Incident Commander, Certification Authority, and constitutional steward as applicable.

Evidence and completion criteria

Completion requires reconciled authoritative state and external effects, satisfied health and outcome gates, closed temporary access, current certification disposition, retained evidence manifest, independent verifier approval, communication to affected Principals, humans, and stakeholders when material, and recorded follow-up ownership.

Examples and anti-patterns

Example: the operator halts a rollout when authority-decision latency rises and evidence gaps appear, preserves the canary, and requests targeted reverification. Anti-pattern: declaring success because processes are live while certification is suspended or state remains unreconciled.

Traceability and review

Constitutional, architecture, engineering, interface, verification, semantic, security/privacy/trust, reliability, and practicability reviews: PASS after final Books VI, VIII, and IX reconciliation. This chapter does not redefine components, contracts, certification authority, or canonical meaning.

Chapter 6 — Capacity, Scaling, Performance, and Resource Governance

Status: FINAL

Responsible role: Capacity Manager

Owner Review items: None

Purpose

Maintain declared service budgets and scale without creating unsafe authority, privacy, or ordering behavior.

Scope

This chapter applies across development, test, staging, recovery, degraded environments, and every declared live-effect environment and approved Reality Boundary stage when the described operational condition exists.

Authority and prerequisites

Books I-IV, IX, and X govern. Book VIII controls verification and certification. Book VI controls the continuing security, privacy, and trust program after reconciliation. Required prerequisites are authenticated operator identity, explicit Authority, exact target identification, an approved change or incident record, current certification, evidence capture, and a tested containment or recovery path.

Normative controls

OPS-CAP-06-01 — Every service

Every service **MUST** have declared capacity, latency, concurrency, queue, storage, and evidence-retention budgets tied to representative workloads.

Applicability: Capacity, Scaling, Performance, and Resource Governance. Responsible: Capacity Manager.

Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.

Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-CAP-06-02 — Scaling

Scaling **MUST** preserve single-owner state, authority checks, ordering scope, rate limits, privacy constraints, and certified topology assumptions.

Applicability: Capacity, Scaling, Performance, and Resource Governance. Responsible: Capacity Manager.
 Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.
 Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-CAP-06-03 — Overload controls

Overload controls **MUST** apply bounded admission, backpressure, prioritization, deferral, degradation, or rejection without silent loss.

Applicability: Capacity, Scaling, Performance, and Resource Governance. Responsible: Capacity Manager.
 Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.
 Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-CAP-06-04 — Capacity exhaustion affecting protected decisions, evidence, authority,

Capacity exhaustion affecting protected decisions, evidence, authority, or recovery **MUST** be treated as an incident and may suspend operation.

Applicability: Capacity, Scaling, Performance, and Resource Governance. Responsible: Capacity Manager.
 Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.
 Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

Mandatory procedure

1. Pre-authorize. Confirm ticket, risk class, acting identity, authority, separation of duties, current certification, approved window, target, and rollback/containment.
2. Baseline. Capture health, state versions, queues, capacity, active transactions, Treaties, alerts, evidence integrity, and human-visible conditions.
3. Validate inputs. Verify signed artifacts/configuration, Book IX compatibility, dependencies, credentials by reference, privacy classification, and expected state.
4. Establish safety. Enable observation, freeze conflicting work, bound blast radius, assign commander and verifier, and announce stop conditions.
5. Execute. Perform the approved action one checkpoint at a time; record operator, timestamp, command or contract, target, result, and evidence digest.
6. Observe. Compare health, outcomes, invariants, authority decisions, privacy signals, queue behavior, and resource use to approved thresholds.

7. Decide. Continue only when success criteria pass and no stop condition fires; otherwise halt, contain, roll back, or enter forward recovery.
8. Reconcile. Resolve state, messages, external effects, in-flight transactions, evidence gaps, and temporary access; do not guess ambiguous reality.
9. Verify. Run targeted Book VIII verification, contract tests, recovery checks, and independent review appropriate to risk.
10. Close. Record final state, limitations, certificate impact, follow-up actions, approvers, evidence manifest, and communication; revoke temporary authority.

Stop conditions

Stop immediately on identity or authority uncertainty, certificate suspension/revocation/expiry, source or target mismatch, integrity failure, unapproved Treaty or Firewall rejection, unexpected protected-state mutation, unbounded queue/resource growth, privacy exposure, ambiguous external effect, failed invariant, or loss of evidence capture.

Rollback, forward recovery, and escalation

Use rollback only when its preconditions are proven and it cannot repeat or hide a committed Reality Boundary effect. Otherwise contain, preserve evidence, reconcile actual state, and use the approved forward-recovery path. Escalate Critical conditions immediately to the Incident Commander, Security Incident Commander, Certification Authority, and constitutional steward as applicable.

Evidence and completion criteria

Completion requires reconciled authoritative state and external effects, satisfied health and outcome gates, closed temporary access, current certification disposition, retained evidence manifest, independent verifier approval, communication to affected Principals, humans, and stakeholders when material, and recorded follow-up ownership.

Examples and anti-patterns

Example: the operator halts a rollout when authority-decision latency rises and evidence gaps appear, preserves the canary, and requests targeted reverification. Anti-pattern: declaring success because processes are live while certification is suspended or state remains unreconciled.

Traceability and review

Constitutional, architecture, engineering, interface, verification, semantic, security/privacy/trust, reliability, and practicability reviews: PASS after final Books VI, VIII, and IX reconciliation. This chapter does not redefine components, contracts, certification authority, or canonical meaning.

Chapter 7 — Monitoring, Alerting, Health, Evidence, and Operational Records

Status: FINAL

Responsible role: Observability Lead

Owner Review items: None

Purpose

Detect material state accurately and preserve attributable, tamper-evident operational evidence under declared privacy, classification, integrity, and retention controls.

Scope

This chapter applies across development, test, staging, recovery, degraded environments, and every declared live-effect environment and approved Reality Boundary stage when the described operational condition exists.

Authority and prerequisites

Books I-IV, IX, and X govern. Book VIII controls verification and certification. Book VI controls the continuing security, privacy, and trust program after reconciliation. Required prerequisites are authenticated operator identity, explicit Authority, exact target identification, an approved change or incident record, current certification, evidence capture, and a tested containment or recovery path.

Normative controls

OPS-OBS-07-01 — Health

Health **MUST** distinguish process liveness, functional readiness, dependency readiness, authority readiness, evidence readiness, and constitutional readiness.

Applicability: Monitoring, Alerting, Health, Evidence, and Operational Records. Responsible: Observability Lead.

Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.

Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-OBS-07-02 — Alerts

Alerts **MUST** name condition, severity, owner, evidence query, immediate safety action, escalation clock, and resolution signal.

Applicability: Monitoring, Alerting, Health, Evidence, and Operational Records. Responsible: Observability Lead. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-OBS-07-03 — Telemetry

Telemetry MUST preserve correlation and causation while minimizing and redacting sensitive content; unrestricted payload logging is prohibited.

Applicability: Monitoring, Alerting, Health, Evidence, and Operational Records. Responsible: Observability Lead. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-OBS-07-04 — Missing, stale, contradictory, or integrity-failed telemetry

Missing, stale, contradictory, or integrity-failed telemetry MUST be represented as unknown and MUST NOT be converted to healthy.

Applicability: Monitoring, Alerting, Health, Evidence, and Operational Records. Responsible: Observability Lead. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

Mandatory procedure

1. Pre-authorize. Confirm ticket, risk class, acting identity, authority, separation of duties, current certification, approved window, target, and rollback/containment.
2. Baseline. Capture health, state versions, queues, capacity, active transactions, Treaties, alerts, evidence integrity, and human-visible conditions.
3. Validate inputs. Verify signed artifacts/configuration, Book IX compatibility, dependencies, credentials by reference, privacy classification, and expected state.
4. Establish safety. Enable observation, freeze conflicting work, bound blast radius, assign commander and verifier, and announce stop conditions.
5. Execute. Perform the approved action one checkpoint at a time; record operator, timestamp, command or contract, target, result, and evidence digest.
6. Observe. Compare health, outcomes, invariants, authority decisions, privacy signals, queue behavior, and resource use to approved thresholds.
7. Decide. Continue only when success criteria pass and no stop condition fires; otherwise halt, contain, roll back, or enter forward recovery.

8. Reconcile. Resolve state, messages, external effects, in-flight transactions, evidence gaps, and temporary access; do not guess ambiguous reality.
9. Verify. Run targeted Book VIII verification, contract tests, recovery checks, and independent review appropriate to risk.
10. Close. Record final state, limitations, certificate impact, follow-up actions, approvers, evidence manifest, and communication; revoke temporary authority.

Stop conditions

Stop immediately on identity or authority uncertainty, certificate suspension/revocation/expiry, source or target mismatch, integrity failure, unapproved Treaty or Firewall rejection, unexpected protected-state mutation, unbounded queue/resource growth, privacy exposure, ambiguous external effect, failed invariant, or loss of evidence capture.

Rollback, forward recovery, and escalation

Use rollback only when its preconditions are proven and it cannot repeat or hide a committed Reality Boundary effect. Otherwise contain, preserve evidence, reconcile actual state, and use the approved forward-recovery path. Escalate Critical conditions immediately to the Incident Commander, Security Incident Commander, Certification Authority, and constitutional steward as applicable.

Evidence and completion criteria

Completion requires reconciled authoritative state and external effects, satisfied health and outcome gates, closed temporary access, current certification disposition, retained evidence manifest, independent verifier approval, communication to affected Principals, humans, and stakeholders when material, and recorded follow-up ownership.

Examples and anti-patterns

Example: the operator halts a rollout when authority-decision latency rises and evidence gaps appear, preserves the canary, and requests targeted reverification. Anti-pattern: declaring success because processes are live while certification is suspended or state remains unreconciled.

Traceability and review

Constitutional, architecture, engineering, interface, verification, semantic, security/privacy/trust, reliability, and practicability reviews: PASS after final Books VI, VIII, and IX reconciliation. This chapter does not redefine components, contracts, certification authority, or canonical meaning.

Chapter 8 — Incident Classification, Command, Response, and Escalation

Status: FINAL

Responsible role: Incident Commander

Owner Review items: None

Purpose

Contain incidents with explicit authority, priority, communications, evidence, and recovery gates.

Scope

This chapter applies across development, test, staging, recovery, degraded environments, and every declared live-effect environment and approved Reality Boundary stage when the described operational condition exists.

Authority and prerequisites

Books I-IV, IX, and X govern. Book VIII controls verification and certification. Book VI controls the continuing security, privacy, and trust program after reconciliation. Required prerequisites are authenticated operator identity, explicit Authority, exact target identification, an approved change or incident record, current certification, evidence capture, and a tested containment or recovery path.

Normative controls

OPS-INC-08-01 — Incidents

Incidents **MUST** be classified by human harm, constitutional impact, authority loss, privacy/security exposure, trust-domain impact, evidence loss, and service effect.

Applicability: Incident Classification, Command, Response, and Escalation. Responsible: Incident Commander.

Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.

Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-INC-08-02 — The Incident Commander

The Incident Commander **MUST** establish scope, authority, priorities, communication cadence, evidence custody, and stop conditions at declaration.

Applicability: Incident Classification, Command, Response, and Escalation. Responsible: Incident Commander.
 Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.
 Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-INC-08-03 — Containment

Containment **MUST** prioritize people, constitutional invariants, authority boundaries, evidence, and reversibility over availability or schedule.

Applicability: Incident Classification, Command, Response, and Escalation. Responsible: Incident Commander.
 Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.
 Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-INC-08-04 — Recovery

Recovery **MUST** not restore reliance until containment, state reconciliation, integrity, targeted verification, and certification consequences are resolved.

Applicability: Incident Classification, Command, Response, and Escalation. Responsible: Incident Commander.
 Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.
 Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

Mandatory procedure

1. Pre-authorize. Confirm ticket, risk class, acting identity, authority, separation of duties, current certification, approved window, target, and rollback/containment.
2. Baseline. Capture health, state versions, queues, capacity, active transactions, Treaties, alerts, evidence integrity, and human-visible conditions.
3. Validate inputs. Verify signed artifacts/configuration, Book IX compatibility, dependencies, credentials by reference, privacy classification, and expected state.
4. Establish safety. Enable observation, freeze conflicting work, bound blast radius, assign commander and verifier, and announce stop conditions.
5. Execute. Perform the approved action one checkpoint at a time; record operator, timestamp, command or contract, target, result, and evidence digest.
6. Observe. Compare health, outcomes, invariants, authority decisions, privacy signals, queue behavior, and resource use to approved thresholds.
7. Decide. Continue only when success criteria pass and no stop condition fires; otherwise halt, contain, roll back, or enter forward recovery.

8. Reconcile. Resolve state, messages, external effects, in-flight transactions, evidence gaps, and temporary access; do not guess ambiguous reality.
9. Verify. Run targeted Book VIII verification, contract tests, recovery checks, and independent review appropriate to risk.
10. Close. Record final state, limitations, certificate impact, follow-up actions, approvers, evidence manifest, and communication; revoke temporary authority.

Stop conditions

Stop immediately on identity or authority uncertainty, certificate suspension/revocation/expiry, source or target mismatch, integrity failure, unapproved Treaty or Firewall rejection, unexpected protected-state mutation, unbounded queue/resource growth, privacy exposure, ambiguous external effect, failed invariant, or loss of evidence capture.

Rollback, forward recovery, and escalation

Use rollback only when its preconditions are proven and it cannot repeat or hide a committed Reality Boundary effect. Otherwise contain, preserve evidence, reconcile actual state, and use the approved forward-recovery path. Escalate Critical conditions immediately to the Incident Commander, Security Incident Commander, Certification Authority, and constitutional steward as applicable.

Evidence and completion criteria

Completion requires reconciled authoritative state and external effects, satisfied health and outcome gates, closed temporary access, current certification disposition, retained evidence manifest, independent verifier approval, communication to affected Principals, humans, and stakeholders when material, and recorded follow-up ownership.

Examples and anti-patterns

Example: the operator halts a rollout when authority-decision latency rises and evidence gaps appear, preserves the canary, and requests targeted reverification. Anti-pattern: declaring success because processes are live while certification is suspended or state remains unreconciled.

Traceability and review

Constitutional, architecture, engineering, interface, verification, semantic, security/privacy/trust, reliability, and practicability reviews: PASS after final Books VI, VIII, and IX reconciliation. This chapter does not redefine components, contracts, certification authority, or canonical meaning.

Chapter 9 — Backup, Restoration, and Data-Recovery Operations

Status: FINAL

Responsible role: Recovery Operator

Owner Review items: None

Purpose

Protect and restore state with integrity, identity continuity, ownership, privacy, and compatibility assurance.

Scope

This chapter applies across development, test, staging, recovery, degraded environments, and every declared live-effect environment and approved Reality Boundary stage when the described operational condition exists.

Authority and prerequisites

Books I-IV, IX, and X govern. Book VIII controls verification and certification. Book VI controls the continuing security, privacy, and trust program after reconciliation. Required prerequisites are authenticated operator identity, explicit Authority, exact target identification, an approved change or incident record, current certification, evidence capture, and a tested containment or recovery path.

Normative controls

OPS-BAK-09-01 — Backup scope

Backup scope MUST cover authoritative state, identity continuity, configuration, policies, evidence, keys by protected reference, schemas, and recovery metadata.

Applicability: Backup, Restoration, and Data-Recovery Operations. Responsible: Recovery Operator. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-BAK-09-02 — Backups

Backups MUST be encrypted, integrity-protected, access-controlled, geographically appropriate, retention-governed, and regularly restorable.

Applicability: Backup, Restoration, and Data-Recovery Operations. Responsible: Recovery Operator. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-BAK-09-03 — Restore

Restore MUST use an isolated target, verified chain of custody, compatible software/schema, authoritative ownership, and reconciliation before promotion.

Applicability: Backup, Restoration, and Data-Recovery Operations. Responsible: Recovery Operator. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-BAK-09-04 — Restoration success

Restoration success MUST prove identity continuity, state integrity, authority validity, privacy obligations, contract compatibility, and recovery objectives.

Applicability: Backup, Restoration, and Data-Recovery Operations. Responsible: Recovery Operator. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

Mandatory procedure

1. Pre-authorize. Confirm ticket, risk class, acting identity, authority, separation of duties, current certification, approved window, target, and rollback/containment.
2. Baseline. Capture health, state versions, queues, capacity, active transactions, Treaties, alerts, evidence integrity, and human-visible conditions.
3. Validate inputs. Verify signed artifacts/configuration, Book IX compatibility, dependencies, credentials by reference, privacy classification, and expected state.
4. Establish safety. Enable observation, freeze conflicting work, bound blast radius, assign commander and verifier, and announce stop conditions.
5. Execute. Perform the approved action one checkpoint at a time; record operator, timestamp, command or contract, target, result, and evidence digest.
6. Observe. Compare health, outcomes, invariants, authority decisions, privacy signals, queue behavior, and resource use to approved thresholds.
7. Decide. Continue only when success criteria pass and no stop condition fires; otherwise halt, contain, roll back, or enter forward recovery.

8. Reconcile. Resolve state, messages, external effects, in-flight transactions, evidence gaps, and temporary access; do not guess ambiguous reality.
9. Verify. Run targeted Book VIII verification, contract tests, recovery checks, and independent review appropriate to risk.
10. Close. Record final state, limitations, certificate impact, follow-up actions, approvers, evidence manifest, and communication; revoke temporary authority.

Stop conditions

Stop immediately on identity or authority uncertainty, certificate suspension/revocation/expiry, source or target mismatch, integrity failure, unapproved Treaty or Firewall rejection, unexpected protected-state mutation, unbounded queue/resource growth, privacy exposure, ambiguous external effect, failed invariant, or loss of evidence capture.

Rollback, forward recovery, and escalation

Use rollback only when its preconditions are proven and it cannot repeat or hide a committed Reality Boundary effect. Otherwise contain, preserve evidence, reconcile actual state, and use the approved forward-recovery path. Escalate Critical conditions immediately to the Incident Commander, Security Incident Commander, Certification Authority, and constitutional steward as applicable.

Evidence and completion criteria

Completion requires reconciled authoritative state and external effects, satisfied health and outcome gates, closed temporary access, current certification disposition, retained evidence manifest, independent verifier approval, communication to affected Principals, humans, and stakeholders when material, and recorded follow-up ownership.

Examples and anti-patterns

Example: the operator halts a rollout when authority-decision latency rises and evidence gaps appear, preserves the canary, and requests targeted reverification. Anti-pattern: declaring success because processes are live while certification is suspended or state remains unreconciled.

Traceability and review

Constitutional, architecture, engineering, interface, verification, semantic, security/privacy/trust, reliability, and practicability reviews: PASS after final Books VI, VIII, and IX reconciliation. This chapter does not redefine components, contracts, certification authority, or canonical meaning.

Chapter 10 — Disaster Recovery, Business Continuity, and Identity Continuity

Status: FINAL

Responsible role: Recovery Coordinator

Owner Review items: None

Purpose

Recover HAL across site or systemic failure without forking identity, authority, or protected state.

Scope

This chapter applies across development, test, staging, recovery, degraded environments, and every declared live-effect environment and approved Reality Boundary stage when the described operational condition exists.

Authority and prerequisites

Books I-IV, IX, and X govern. Book VIII controls verification and certification. Book VI controls the continuing security, privacy, and trust program after reconciliation. Required prerequisites are authenticated operator identity, explicit Authority, exact target identification, an approved change or incident record, current certification, evidence capture, and a tested containment or recovery path.

Normative controls

OPS-DR-10-01 — Disaster declarations

Disaster declarations MUST identify affected identities, state domains, trust boundaries, certifications, Treaties, and Reality Boundary actions.

Applicability: Disaster Recovery, Business Continuity, and Identity Continuity. Responsible: Recovery Coordinator.

Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.

Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-DR-10-02 — Failover

Failover MUST preserve one authoritative identity and mutation owner per state domain and MUST fence superseded writers before admission.

Applicability: Disaster Recovery, Business Continuity, and Identity Continuity. Responsible: Recovery Coordinator. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-DR-10-03 — Continuity modes

Continuity modes MUST state unavailable capabilities, reduced assurances, time bounds, human notification, and prohibited protected actions.

Applicability: Disaster Recovery, Business Continuity, and Identity Continuity. Responsible: Recovery Coordinator. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-DR-10-04 — Return to primary operation

Return to primary operation MUST reconcile state and effects, close gaps, verify invariants, revoke temporary access, and obtain recovery admission.

Applicability: Disaster Recovery, Business Continuity, and Identity Continuity. Responsible: Recovery Coordinator. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

Mandatory procedure

1. Pre-authorize. Confirm ticket, risk class, acting identity, authority, separation of duties, current certification, approved window, target, and rollback/containment.
2. Baseline. Capture health, state versions, queues, capacity, active transactions, Treaties, alerts, evidence integrity, and human-visible conditions.
3. Validate inputs. Verify signed artifacts/configuration, Book IX compatibility, dependencies, credentials by reference, privacy classification, and expected state.
4. Establish safety. Enable observation, freeze conflicting work, bound blast radius, assign commander and verifier, and announce stop conditions.
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8. Reconcile. Resolve state, messages, external effects, in-flight transactions, evidence gaps, and temporary access; do not guess ambiguous reality.
9. Verify. Run targeted Book VIII verification, contract tests, recovery checks, and independent review appropriate to risk.
10. Close. Record final state, limitations, certificate impact, follow-up actions, approvers, evidence manifest, and communication; revoke temporary authority.

Stop conditions

Stop immediately on identity or authority uncertainty, certificate suspension/revocation/expiry, source or target mismatch, integrity failure, unapproved Treaty or Firewall rejection, unexpected protected-state mutation, unbounded queue/resource growth, privacy exposure, ambiguous external effect, failed invariant, or loss of evidence capture.

Rollback, forward recovery, and escalation

Use rollback only when its preconditions are proven and it cannot repeat or hide a committed Reality Boundary effect. Otherwise contain, preserve evidence, reconcile actual state, and use the approved forward-recovery path. Escalate Critical conditions immediately to the Incident Commander, Security Incident Commander, Certification Authority, and constitutional steward as applicable.

Evidence and completion criteria

Completion requires reconciled authoritative state and external effects, satisfied health and outcome gates, closed temporary access, current certification disposition, retained evidence manifest, independent verifier approval, communication to affected Principals, humans, and stakeholders when material, and recorded follow-up ownership.

Examples and anti-patterns

Example: the operator halts a rollout when authority-decision latency rises and evidence gaps appear, preserves the canary, and requests targeted reverification. Anti-pattern: declaring success because processes are live while certification is suspended or state remains unreconciled.

Traceability and review

Constitutional, architecture, engineering, interface, verification, semantic, security/privacy/trust, reliability, and practicability reviews: PASS after final Books VI, VIII, and IX reconciliation. This chapter does not redefine components, contracts, certification authority, or canonical meaning.

Chapter 11 — Maintenance, Upgrade, Patch, Deprecation, and Removal

Status: FINAL

Responsible role: Service Owner

Owner Review items: None

Purpose

Perform maintenance and lifecycle change through risk classification, compatibility, verification, and rollback.

Scope

This chapter applies across development, test, staging, recovery, degraded environments, and every declared live-effect environment and approved Reality Boundary stage when the described operational condition exists.

Authority and prerequisites

Books I-IV, IX, and X govern. Book VIII controls verification and certification. Book VI controls the continuing security, privacy, and trust program after reconciliation. Required prerequisites are authenticated operator identity, explicit Authority, exact target identification, an approved change or incident record, current certification, evidence capture, and a tested containment or recovery path.

Normative controls

OPS-MNT-11-01 — Maintenance

Maintenance MUST declare affected components, contracts, state, certificates, dependencies, observation windows, and rollback or forward-recovery plan.

Applicability: Maintenance, Upgrade, Patch, Deprecation, and Removal. Responsible: Service Owner. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-MNT-11-02 — Patches

Patches MUST pass risk-proportionate verification and current security disposition before admission to the declared live-effect environment and approved Reality Boundary stage.

Applicability: Maintenance, Upgrade, Patch, Deprecation, and Removal. Responsible: Service Owner. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-MNT-11-03 — Deprecation

Deprecation **MUST** preserve supported coexistence, consumer notice, migration evidence, removal criteria, and rollback until the approved end date.

Applicability: Maintenance, Upgrade, Patch, Deprecation, and Removal. Responsible: Service Owner. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-MNT-11-04 — Removal

Removal **MUST** verify no authorized consumer, retained evidence, recovery path, Treaty, certificate, or state obligation depends on the target.

Applicability: Maintenance, Upgrade, Patch, Deprecation, and Removal. Responsible: Service Owner. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

Mandatory procedure

1. Pre-authorize. Confirm ticket, risk class, acting identity, authority, separation of duties, current certification, approved window, target, and rollback/containment.
2. Baseline. Capture health, state versions, queues, capacity, active transactions, Treaties, alerts, evidence integrity, and human-visible conditions.
3. Validate inputs. Verify signed artifacts/configuration, Book IX compatibility, dependencies, credentials by reference, privacy classification, and expected state.
4. Establish safety. Enable observation, freeze conflicting work, bound blast radius, assign commander and verifier, and announce stop conditions.
5. Execute. Perform the approved action one checkpoint at a time; record operator, timestamp, command or contract, target, result, and evidence digest.
6. Observe. Compare health, outcomes, invariants, authority decisions, privacy signals, queue behavior, and resource use to approved thresholds.
7. Decide. Continue only when success criteria pass and no stop condition fires; otherwise halt, contain, roll back, or enter forward recovery.

8. Reconcile. Resolve state, messages, external effects, in-flight transactions, evidence gaps, and temporary access; do not guess ambiguous reality.
9. Verify. Run targeted Book VIII verification, contract tests, recovery checks, and independent review appropriate to risk.
10. Close. Record final state, limitations, certificate impact, follow-up actions, approvers, evidence manifest, and communication; revoke temporary authority.

Stop conditions

Stop immediately on identity or authority uncertainty, certificate suspension/revocation/expiry, source or target mismatch, integrity failure, unapproved Treaty or Firewall rejection, unexpected protected-state mutation, unbounded queue/resource growth, privacy exposure, ambiguous external effect, failed invariant, or loss of evidence capture.

Rollback, forward recovery, and escalation

Use rollback only when its preconditions are proven and it cannot repeat or hide a committed Reality Boundary effect. Otherwise contain, preserve evidence, reconcile actual state, and use the approved forward-recovery path. Escalate Critical conditions immediately to the Incident Commander, Security Incident Commander, Certification Authority, and constitutional steward as applicable.

Evidence and completion criteria

Completion requires reconciled authoritative state and external effects, satisfied health and outcome gates, closed temporary access, current certification disposition, retained evidence manifest, independent verifier approval, communication to affected Principals, humans, and stakeholders when material, and recorded follow-up ownership.

Examples and anti-patterns

Example: the operator halts a rollout when authority-decision latency rises and evidence gaps appear, preserves the canary, and requests targeted reverification. Anti-pattern: declaring success because processes are live while certification is suspended or state remains unreconciled.

Traceability and review

Constitutional, architecture, engineering, interface, verification, semantic, security/privacy/trust, reliability, and practicability reviews: PASS after final Books VI, VIII, and IX reconciliation. This chapter does not redefine components, contracts, certification authority, or canonical meaning.

Chapter 12 — Database, State, and Contract Migration Operations

Status: FINAL

Responsible role: Migration Lead

Owner Review items: None

Purpose

Migrate authoritative state and contracts with single-writer ownership, reconciliation, coexistence, and evidence.

Scope

This chapter applies across development, test, staging, recovery, degraded environments, and every declared live-effect environment and approved Reality Boundary stage when the described operational condition exists.

Authority and prerequisites

Books I-IV, IX, and X govern. Book VIII controls verification and certification. Book VI controls the continuing security, privacy, and trust program after reconciliation. Required prerequisites are authenticated operator identity, explicit Authority, exact target identification, an approved change or incident record, current certification, evidence capture, and a tested containment or recovery path.

Normative controls

OPS-MIG-12-01 — Migration

Migration MUST identify the sole mutation owner, source and target schemas, invariants, transformation, checkpoints, reconciliation, and abort threshold.

Applicability: Database, State, and Contract Migration Operations. Responsible: Migration Lead. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-MIG-12-02 — Dual writes are prohibited unless an approved consistency protocol names

Dual writes are prohibited unless an approved consistency protocol names the authority, ordering, conflict, replay, and recovery semantics.

Applicability: Database, State, and Contract Migration Operations. Responsible: Migration Lead. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-MIG-12-03 — Migration execution

Migration execution **MUST** be idempotent or checkpointed and **MUST** preserve provenance, classification, retention, and evidence linkage.

Applicability: Database, State, and Contract Migration Operations. Responsible: Migration Lead. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-MIG-12-04 — Cutover

Cutover **MUST** require validated counts and invariants, consumer compatibility, quiescence or ordered handoff, and explicit state-owner acceptance.

Applicability: Database, State, and Contract Migration Operations. Responsible: Migration Lead. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

Mandatory procedure

1. Pre-authorize. Confirm ticket, risk class, acting identity, authority, separation of duties, current certification, approved window, target, and rollback/containment.
2. Baseline. Capture health, state versions, queues, capacity, active transactions, Treaties, alerts, evidence integrity, and human-visible conditions.
3. Validate inputs. Verify signed artifacts/configuration, Book IX compatibility, dependencies, credentials by reference, privacy classification, and expected state.
4. Establish safety. Enable observation, freeze conflicting work, bound blast radius, assign commander and verifier, and announce stop conditions.
5. Execute. Perform the approved action one checkpoint at a time; record operator, timestamp, command or contract, target, result, and evidence digest.
6. Observe. Compare health, outcomes, invariants, authority decisions, privacy signals, queue behavior, and resource use to approved thresholds.
7. Decide. Continue only when success criteria pass and no stop condition fires; otherwise halt, contain, roll back, or enter forward recovery.

8. Reconcile. Resolve state, messages, external effects, in-flight transactions, evidence gaps, and temporary access; do not guess ambiguous reality.
9. Verify. Run targeted Book VIII verification, contract tests, recovery checks, and independent review appropriate to risk.
10. Close. Record final state, limitations, certificate impact, follow-up actions, approvers, evidence manifest, and communication; revoke temporary authority.

Stop conditions

Stop immediately on identity or authority uncertainty, certificate suspension/revocation/expiry, source or target mismatch, integrity failure, unapproved Treaty or Firewall rejection, unexpected protected-state mutation, unbounded queue/resource growth, privacy exposure, ambiguous external effect, failed invariant, or loss of evidence capture.

Rollback, forward recovery, and escalation

Use rollback only when its preconditions are proven and it cannot repeat or hide a committed Reality Boundary effect. Otherwise contain, preserve evidence, reconcile actual state, and use the approved forward-recovery path. Escalate Critical conditions immediately to the Incident Commander, Security Incident Commander, Certification Authority, and constitutional steward as applicable.

Evidence and completion criteria

Completion requires reconciled authoritative state and external effects, satisfied health and outcome gates, closed temporary access, current certification disposition, retained evidence manifest, independent verifier approval, communication to affected Principals, humans, and stakeholders when material, and recorded follow-up ownership.

Examples and anti-patterns

Example: the operator halts a rollout when authority-decision latency rises and evidence gaps appear, preserves the canary, and requests targeted reverification. Anti-pattern: declaring success because processes are live while certification is suspended or state remains unreconciled.

Traceability and review

Constitutional, architecture, engineering, interface, verification, semantic, security/privacy/trust, reliability, and practicability reviews: PASS after final Books VI, VIII, and IX reconciliation. This chapter does not redefine components, contracts, certification authority, or canonical meaning.

Chapter 13 — External Domains, Constitutional Firewall, and Treaty Operations

Status: FINAL

Responsible role: Trust Operator

Owner Review items: None

Purpose

Operate external exchanges only through active applicable Treaties and Constitutional Firewall admission.

Scope

This chapter applies across development, test, staging, recovery, degraded environments, and every declared live-effect environment and approved Reality Boundary stage when the described operational condition exists.

Authority and prerequisites

Books I-IV, IX, and X govern. Book VIII controls verification and certification. Book VI controls the continuing security, privacy, and trust program after reconciliation. Required prerequisites are authenticated operator identity, explicit Authority, exact target identification, an approved change or incident record, current certification, evidence capture, and a tested containment or recovery path.

Normative controls

OPS-TRT-13-01 — No external exchange may begin without authenticated domain identity, an

No external exchange may begin without authenticated domain identity, an active applicable Treaty, current authority, and Constitutional Firewall admission.

Applicability: External Domains, Constitutional Firewall, and Treaty Operations. Responsible: Trust Operator.

Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.

Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-TRT-13-02 — Treaty activation

Treaty activation **MUST** verify exact approved text, parties, purpose, capability, data, duration, audit, revocation, and Firewall policy binding.

Applicability: External Domains, Constitutional Firewall, and Treaty Operations. Responsible: Trust Operator.

Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.

Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-TRT-13-03 — Treaty suspension, expiry, revocation, drift, or external compromise

Treaty suspension, expiry, revocation, drift, or external compromise **MUST** stop new exchange and safely contain or reconcile in-flight work.

Applicability: External Domains, Constitutional Firewall, and Treaty Operations. Responsible: Trust Operator.

Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.

Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-TRT-13-04 — Firewall bypass, direct external credentials, and ungoverned side channel

Firewall bypass, direct external credentials, and ungoverned side channels are prohibited even during incidents or degraded operation.

Applicability: External Domains, Constitutional Firewall, and Treaty Operations. Responsible: Trust Operator.

Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.

Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

Mandatory procedure

1. Pre-authorize. Confirm ticket, risk class, acting identity, authority, separation of duties, current certification, approved window, target, and rollback/containment.
2. Baseline. Capture health, state versions, queues, capacity, active transactions, Treaties, alerts, evidence integrity, and human-visible conditions.
3. Validate inputs. Verify signed artifacts/configuration, Book IX compatibility, dependencies, credentials by reference, privacy classification, and expected state.
4. Establish safety. Enable observation, freeze conflicting work, bound blast radius, assign commander and verifier, and announce stop conditions.

5. Execute. Perform the approved action one checkpoint at a time; record operator, timestamp, command or contract, target, result, and evidence digest.
6. Observe. Compare health, outcomes, invariants, authority decisions, privacy signals, queue behavior, and resource use to approved thresholds.
7. Decide. Continue only when success criteria pass and no stop condition fires; otherwise halt, contain, roll back, or enter forward recovery.
8. Reconcile. Resolve state, messages, external effects, in-flight transactions, evidence gaps, and temporary access; do not guess ambiguous reality.
9. Verify. Run targeted Book VIII verification, contract tests, recovery checks, and independent review appropriate to risk.
10. Close. Record final state, limitations, certificate impact, follow-up actions, approvers, evidence manifest, and communication; revoke temporary authority.

Stop conditions

Stop immediately on identity or authority uncertainty, certificate suspension/revocation/expiry, source or target mismatch, integrity failure, unapproved Treaty or Firewall rejection, unexpected protected-state mutation, unbounded queue/resource growth, privacy exposure, ambiguous external effect, failed invariant, or loss of evidence capture.

Rollback, forward recovery, and escalation

Use rollback only when its preconditions are proven and it cannot repeat or hide a committed Reality Boundary effect. Otherwise contain, preserve evidence, reconcile actual state, and use the approved forward-recovery path. Escalate Critical conditions immediately to the Incident Commander, Security Incident Commander, Certification Authority, and constitutional steward as applicable.

Evidence and completion criteria

Completion requires reconciled authoritative state and external effects, satisfied health and outcome gates, closed temporary access, current certification disposition, retained evidence manifest, independent verifier approval, communication to affected Principals, humans, and stakeholders when material, and recorded follow-up ownership.

Examples and anti-patterns

Example: the operator halts a rollout when authority-decision latency rises and evidence gaps appear, preserves the canary, and requests targeted reverification. Anti-pattern: declaring success because processes are live while certification is suspended or state remains unreconciled.

Traceability and review

Constitutional, architecture, engineering, interface, verification, semantic, security/privacy/trust, reliability, and practicability reviews: PASS after final Books VI, VIII, and IX reconciliation. This chapter does not redefine components, contracts, certification authority, or canonical meaning.

Chapter 14 — Degraded Modes, Failure Containment, and Service Restoration

Status: FINAL

Responsible role: Runtime Operator

Owner Review items: None

Purpose

Enter, operate, and exit degraded modes safely with disclosed limitations and bounded authority.

Scope

This chapter applies across development, test, staging, recovery, degraded environments, and every declared live-effect environment and approved Reality Boundary stage when the described operational condition exists.

Authority and prerequisites

Books I-IV, IX, and X govern. Book VIII controls verification and certification. Book VI controls the continuing security, privacy, and trust program after reconciliation. Required prerequisites are authenticated operator identity, explicit Authority, exact target identification, an approved change or incident record, current certification, evidence capture, and a tested containment or recovery path.

Normative controls

OPS-DEG-14-01 — Every degraded mode

Every degraded mode **MUST** be predefined with entry triggers, available and prohibited capabilities, authority ceiling, evidence, time bound, and exit criteria.

Applicability: Degraded Modes, Failure Containment, and Service Restoration. Responsible: Runtime Operator.

Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.

Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-DEG-14-02 — Degradation

Degradation MUST fail closed for uncertain authority, Treaty, privacy, integrity, or Reality Boundary state unless a higher rule explicitly requires a safer human-protective action.

Applicability: Degraded Modes, Failure Containment, and Service Restoration. Responsible: Runtime Operator.

Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.

Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-DEG-14-03 — Users and operators

Users and operators MUST receive accurate limitation and uncertainty disclosure without claiming unavailable capability or evidence.

Applicability: Degraded Modes, Failure Containment, and Service Restoration. Responsible: Runtime Operator.

Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.

Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-DEG-14-04 — Restoration

Restoration MUST verify repaired dependencies, reconcile queued and in-flight work, clear temporary controls, and re-establish required certification.

Applicability: Degraded Modes, Failure Containment, and Service Restoration. Responsible: Runtime Operator.

Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.

Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

Mandatory procedure

1. Pre-authorize. Confirm ticket, risk class, acting identity, authority, separation of duties, current certification, approved window, target, and rollback/containment.
2. Baseline. Capture health, state versions, queues, capacity, active transactions, Treaties, alerts, evidence integrity, and human-visible conditions.
3. Validate inputs. Verify signed artifacts/configuration, Book IX compatibility, dependencies, credentials by reference, privacy classification, and expected state.
4. Establish safety. Enable observation, freeze conflicting work, bound blast radius, assign commander and verifier, and announce stop conditions.
5. Execute. Perform the approved action one checkpoint at a time; record operator, timestamp, command or contract, target, result, and evidence digest.

6. Observe. Compare health, outcomes, invariants, authority decisions, privacy signals, queue behavior, and resource use to approved thresholds.
7. Decide. Continue only when success criteria pass and no stop condition fires; otherwise halt, contain, roll back, or enter forward recovery.
8. Reconcile. Resolve state, messages, external effects, in-flight transactions, evidence gaps, and temporary access; do not guess ambiguous reality.
9. Verify. Run targeted Book VIII verification, contract tests, recovery checks, and independent review appropriate to risk.
10. Close. Record final state, limitations, certificate impact, follow-up actions, approvers, evidence manifest, and communication; revoke temporary authority.

Stop conditions

Stop immediately on identity or authority uncertainty, certificate suspension/revocation/expiry, source or target mismatch, integrity failure, unapproved Treaty or Firewall rejection, unexpected protected-state mutation, unbounded queue/resource growth, privacy exposure, ambiguous external effect, failed invariant, or loss of evidence capture.

Rollback, forward recovery, and escalation

Use rollback only when its preconditions are proven and it cannot repeat or hide a committed Reality Boundary effect. Otherwise contain, preserve evidence, reconcile actual state, and use the approved forward-recovery path. Escalate Critical conditions immediately to the Incident Commander, Security Incident Commander, Certification Authority, and constitutional steward as applicable.

Evidence and completion criteria

Completion requires reconciled authoritative state and external effects, satisfied health and outcome gates, closed temporary access, current certification disposition, retained evidence manifest, independent verifier approval, communication to affected Principals, humans, and stakeholders when material, and recorded follow-up ownership.

Examples and anti-patterns

Example: the operator halts a rollout when authority-decision latency rises and evidence gaps appear, preserves the canary, and requests targeted reverification. Anti-pattern: declaring success because processes are live while certification is suspended or state remains unreconciled.

Traceability and review

Constitutional, architecture, engineering, interface, verification, semantic, security/privacy/trust, reliability, and practicability reviews: PASS after final Books VI, VIII, and IX reconciliation. This chapter does not redefine components, contracts, certification authority, or canonical meaning.

Chapter 15 — Emergency Changes and Constitutional Shutdown

Status: FINAL

Responsible role: Incident Commander

Owner Review items: None

Purpose

Execute emergency change or constitutional shutdown under narrow, attributable, time-bounded authority.

Scope

This chapter applies across development, test, staging, recovery, degraded environments, and every declared live-effect environment and approved Reality Boundary stage when the described operational condition exists.

Authority and prerequisites

Books I-IV, IX, and X govern. Book VIII controls verification and certification. Book VI controls the continuing security, privacy, and trust program after reconciliation. Required prerequisites are authenticated operator identity, explicit Authority, exact target identification, an approved change or incident record, current certification, evidence capture, and a tested containment or recovery path.

Normative controls

OPS-EMG-15-01 — Emergency change authority

Emergency change authority **MUST** be narrow, attributable, time-bounded, independently reviewed, and unable to waive constitutional invariants.

Applicability: Emergency Changes and Constitutional Shutdown. Responsible: Incident Commander. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-EMG-15-02 — Emergency changes

Emergency changes **MUST** preserve evidence, predefine containment and rollback, minimize scope, and receive retrospective verification before continued use.

Applicability: Emergency Changes and Constitutional Shutdown. Responsible: Incident Commander. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-EMG-15-03 — Constitutional shutdown

Constitutional shutdown MUST stop protected action admission, external exchange, hazardous mutation, and new delegation while preserving harm-bounded human communication and evidence.

Applicability: Emergency Changes and Constitutional Shutdown. Responsible: Incident Commander. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-EMG-15-04 — Restart after constitutional shutdown

Restart after constitutional shutdown MUST require root-cause containment, integrity and identity Verification results, Book VIII reverification, and explicit recovery admission.

Applicability: Emergency Changes and Constitutional Shutdown. Responsible: Incident Commander. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

Mandatory procedure

1. Pre-authorize. Confirm ticket, risk class, acting identity, authority, separation of duties, current certification, approved window, target, and rollback/containment.
2. Baseline. Capture health, state versions, queues, capacity, active transactions, Treaties, alerts, evidence integrity, and human-visible conditions.
3. Validate inputs. Verify signed artifacts/configuration, Book IX compatibility, dependencies, credentials by reference, privacy classification, and expected state.
4. Establish safety. Enable observation, freeze conflicting work, bound blast radius, assign commander and verifier, and announce stop conditions.
5. Execute. Perform the approved action one checkpoint at a time; record operator, timestamp, command or contract, target, result, and evidence digest.
6. Observe. Compare health, outcomes, invariants, authority decisions, privacy signals, queue behavior, and resource use to approved thresholds.
7. Decide. Continue only when success criteria pass and no stop condition fires; otherwise halt, contain, roll back, or enter forward recovery.

8. Reconcile. Resolve state, messages, external effects, in-flight transactions, evidence gaps, and temporary access; do not guess ambiguous reality.
9. Verify. Run targeted Book VIII verification, contract tests, recovery checks, and independent review appropriate to risk.
10. Close. Record final state, limitations, certificate impact, follow-up actions, approvers, evidence manifest, and communication; revoke temporary authority.

Stop conditions

Stop immediately on identity or authority uncertainty, certificate suspension/revocation/expiry, source or target mismatch, integrity failure, unapproved Treaty or Firewall rejection, unexpected protected-state mutation, unbounded queue/resource growth, privacy exposure, ambiguous external effect, failed invariant, or loss of evidence capture.

Rollback, forward recovery, and escalation

Use rollback only when its preconditions are proven and it cannot repeat or hide a committed Reality Boundary effect. Otherwise contain, preserve evidence, reconcile actual state, and use the approved forward-recovery path. Escalate Critical conditions immediately to the Incident Commander, Security Incident Commander, Certification Authority, and constitutional steward as applicable.

Evidence and completion criteria

Completion requires reconciled authoritative state and external effects, satisfied health and outcome gates, closed temporary access, current certification disposition, retained evidence manifest, independent verifier approval, communication to affected Principals, humans, and stakeholders when material, and recorded follow-up ownership.

Examples and anti-patterns

Example: the operator halts a rollout when authority-decision latency rises and evidence gaps appear, preserves the canary, and requests targeted reverification. Anti-pattern: declaring success because processes are live while certification is suspended or state remains unreconciled.

Traceability and review

Constitutional, architecture, engineering, interface, verification, semantic, security/privacy/trust, reliability, and practicability reviews: PASS after final Books VI, VIII, and IX reconciliation. This chapter does not redefine components, contracts, certification authority, or canonical meaning.

Chapter 16 — Post-Incident Review, Corrective Action, and Recertification

Status: FINAL

Responsible role: Problem Manager

Owner Review items: None

Purpose

Turn incidents into verified corrections while preserving negative evidence and certification consequences.

Scope

This chapter applies across development, test, staging, recovery, degraded environments, and every declared live-effect environment and approved Reality Boundary stage when the described operational condition exists.

Authority and prerequisites

Books I-IV, IX, and X govern. Book VIII controls verification and certification. Book VI controls the continuing security, privacy, and trust program after reconciliation. Required prerequisites are authenticated operator identity, explicit Authority, exact target identification, an approved change or incident record, current certification, evidence capture, and a tested containment or recovery path.

Normative controls

OPS-PIR-16-01 — Every material incident

Every material incident **MUST** receive a blameless evidence-based review covering timeline, authority, decisions, contributing conditions, controls, harms, and outcomes.

Applicability: Post-Incident Review, Corrective Action, and Recertification. Responsible: Problem Manager.

Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.

Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-PIR-16-02 — Corrective actions

Corrective actions **MUST** have owners, risk, due dates, verification methods, evidence, and closure authority; administrative closure without an admitted Evidence Object and accepted Verification result is prohibited.

Applicability: Post-Incident Review, Corrective Action, and Recertification. Responsible: Problem Manager.
 Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.
 Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-PIR-16-03 — Incident impact on claims and certificates

Incident impact on claims and certificates **MUST** be assessed immediately; affected reliance **MUST** remain suspended until targeted reverification passes.

Applicability: Post-Incident Review, Corrective Action, and Recertification. Responsible: Problem Manager.
 Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.
 Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-PIR-16-04 — Lessons

Lessons **MUST** update controls, runbooks, alerts, tests, capacity models, training, and risk registers without erasing negative evidence.

Applicability: Post-Incident Review, Corrective Action, and Recertification. Responsible: Problem Manager.
 Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.
 Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

Mandatory procedure

1. Pre-authorize. Confirm ticket, risk class, acting identity, authority, separation of duties, current certification, approved window, target, and rollback/containment.
2. Baseline. Capture health, state versions, queues, capacity, active transactions, Treaties, alerts, evidence integrity, and human-visible conditions.
3. Validate inputs. Verify signed artifacts/configuration, Book IX compatibility, dependencies, credentials by reference, privacy classification, and expected state.
4. Establish safety. Enable observation, freeze conflicting work, bound blast radius, assign commander and verifier, and announce stop conditions.
5. Execute. Perform the approved action one checkpoint at a time; record operator, timestamp, command or contract, target, result, and evidence digest.
6. Observe. Compare health, outcomes, invariants, authority decisions, privacy signals, queue behavior, and resource use to approved thresholds.
7. Decide. Continue only when success criteria pass and no stop condition fires; otherwise halt, contain, roll back, or enter forward recovery.

8. Reconcile. Resolve state, messages, external effects, in-flight transactions, evidence gaps, and temporary access; do not guess ambiguous reality.
9. Verify. Run targeted Book VIII verification, contract tests, recovery checks, and independent review appropriate to risk.
10. Close. Record final state, limitations, certificate impact, follow-up actions, approvers, evidence manifest, and communication; revoke temporary authority.

Stop conditions

Stop immediately on identity or authority uncertainty, certificate suspension/revocation/expiry, source or target mismatch, integrity failure, unapproved Treaty or Firewall rejection, unexpected protected-state mutation, unbounded queue/resource growth, privacy exposure, ambiguous external effect, failed invariant, or loss of evidence capture.

Rollback, forward recovery, and escalation

Use rollback only when its preconditions are proven and it cannot repeat or hide a committed Reality Boundary effect. Otherwise contain, preserve evidence, reconcile actual state, and use the approved forward-recovery path. Escalate Critical conditions immediately to the Incident Commander, Security Incident Commander, Certification Authority, and constitutional steward as applicable.

Evidence and completion criteria

Completion requires reconciled authoritative state and external effects, satisfied health and outcome gates, closed temporary access, current certification disposition, retained evidence manifest, independent verifier approval, communication to affected Principals, humans, and stakeholders when material, and recorded follow-up ownership.

Examples and anti-patterns

Example: the operator halts a rollout when authority-decision latency rises and evidence gaps appear, preserves the canary, and requests targeted reverification. Anti-pattern: declaring success because processes are live while certification is suspended or state remains unreconciled.

Traceability and review

Constitutional, architecture, engineering, interface, verification, semantic, security/privacy/trust, reliability, and practicability reviews: PASS after final Books VI, VIII, and IX reconciliation. This chapter does not redefine components, contracts, certification authority, or canonical meaning.

Chapter 17 — Operational Evidence Retention, Audit, and Reporting

Status: FINAL

Responsible role: Evidence Custodian

Owner Review items: None

Purpose

Retain, protect, disclose, and dispose operational evidence according to purpose, classification, and certification needs.

Scope

This chapter applies across development, test, staging, recovery, degraded environments, and every declared live-effect environment and approved Reality Boundary stage when the described operational condition exists.

Authority and prerequisites

Books I-IV, IX, and X govern. Book VIII controls verification and certification. Book VI controls the continuing security, privacy, and trust program after reconciliation. Required prerequisites are authenticated operator identity, explicit Authority, exact target identification, an approved change or incident record, current certification, evidence capture, and a tested containment or recovery path.

Normative controls

OPS-EVD-17-01 — Operational evidence

Operational evidence **MUST** be attributable, time-bound, integrity-protected, classified, purpose-linked, access-controlled, and retention-governed.

Applicability: Operational Evidence Retention, Audit, and Reporting. Responsible: Evidence Custodian.

Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.

Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-EVD-17-02 — Evidence retention

Evidence retention **MUST** satisfy constitutional, certification, security, privacy, incident, Treaty, and recovery needs without indefinite collection by default.

Applicability: Operational Evidence Retention, Audit, and Reporting. Responsible: Evidence Custodian.
 Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.
 Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-EVD-17-03 — Audit access and export

Audit access and export **MUST** be separately authorized, minimized, logged, and subject to Treaty and Firewall controls across trust domains.

Applicability: Operational Evidence Retention, Audit, and Reporting. Responsible: Evidence Custodian.
 Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.
 Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-EVD-17-04 — Disposition

Disposition **MUST** verify holds, dependent claims, active incidents, recovery needs, and deletion evidence before irreversible destruction.

Applicability: Operational Evidence Retention, Audit, and Reporting. Responsible: Evidence Custodian.
 Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable.
 Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

Mandatory procedure

1. Pre-authorize. Confirm ticket, risk class, acting identity, authority, separation of duties, current certification, approved window, target, and rollback/containment.
2. Baseline. Capture health, state versions, queues, capacity, active transactions, Treaties, alerts, evidence integrity, and human-visible conditions.
3. Validate inputs. Verify signed artifacts/configuration, Book IX compatibility, dependencies, credentials by reference, privacy classification, and expected state.
4. Establish safety. Enable observation, freeze conflicting work, bound blast radius, assign commander and verifier, and announce stop conditions.
5. Execute. Perform the approved action one checkpoint at a time; record operator, timestamp, command or contract, target, result, and evidence digest.
6. Observe. Compare health, outcomes, invariants, authority decisions, privacy signals, queue behavior, and resource use to approved thresholds.
7. Decide. Continue only when success criteria pass and no stop condition fires; otherwise halt, contain, roll back, or enter forward recovery.

8. Reconcile. Resolve state, messages, external effects, in-flight transactions, evidence gaps, and temporary access; do not guess ambiguous reality.
9. Verify. Run targeted Book VIII verification, contract tests, recovery checks, and independent review appropriate to risk.
10. Close. Record final state, limitations, certificate impact, follow-up actions, approvers, evidence manifest, and communication; revoke temporary authority.

Stop conditions

Stop immediately on identity or authority uncertainty, certificate suspension/revocation/expiry, source or target mismatch, integrity failure, unapproved Treaty or Firewall rejection, unexpected protected-state mutation, unbounded queue/resource growth, privacy exposure, ambiguous external effect, failed invariant, or loss of evidence capture.

Rollback, forward recovery, and escalation

Use rollback only when its preconditions are proven and it cannot repeat or hide a committed Reality Boundary effect. Otherwise contain, preserve evidence, reconcile actual state, and use the approved forward-recovery path. Escalate Critical conditions immediately to the Incident Commander, Security Incident Commander, Certification Authority, and constitutional steward as applicable.

Evidence and completion criteria

Completion requires reconciled authoritative state and external effects, satisfied health and outcome gates, closed temporary access, current certification disposition, retained evidence manifest, independent verifier approval, communication to affected Principals, humans, and stakeholders when material, and recorded follow-up ownership.

Examples and anti-patterns

Example: the operator halts a rollout when authority-decision latency rises and evidence gaps appear, preserves the canary, and requests targeted reverification. Anti-pattern: declaring success because processes are live while certification is suspended or state remains unreconciled.

Traceability and review

Constitutional, architecture, engineering, interface, verification, semantic, security/privacy/trust, reliability, and practicability reviews: PASS after final Books VI, VIII, and IX reconciliation. This chapter does not redefine components, contracts, certification authority, or canonical meaning.

Chapter 18 — Operational Readiness, Verification, Certification, and Continuous Assurance

Status: FINAL

Responsible role: Operations Assurance Lead

Owner Review items: None

Purpose

Prove readiness and keep operational reliance aligned with Book VIII certification, suspension, and revocation states.

Scope

This chapter applies across development, test, staging, recovery, degraded environments, and every declared live-effect environment and approved Reality Boundary stage when the described operational condition exists.

Authority and prerequisites

Books I-IV, IX, and X govern. Book VIII controls verification and certification. Book VI controls the continuing security, privacy, and trust program after reconciliation. Required prerequisites are authenticated operator identity, explicit Authority, exact target identification, an approved change or incident record, current certification, evidence capture, and a tested containment or recovery path.

Normative controls

OPS-RDY-18-01 — Operational readiness

Operational readiness **MUST** prove people, authority, dependencies, capacity, monitoring, response, backup, recovery, security, privacy, contracts, and evidence.

Applicability: Operational Readiness, Verification, Certification, and Continuous Assurance. Responsible: Operations Assurance Lead. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-RDY-18-02 — Runtime admission

Runtime admission **MUST** check current Book VIII certification scope and **MUST** fail closed on suspension, revocation, expiry, missing evidence, or material drift.

Applicability: Operational Readiness, Verification, Certification, and Continuous Assurance. Responsible: Operations Assurance Lead. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: High. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-RDY-18-03 — Continuous assurance

Continuous assurance **MUST** correlate changes, incidents, drift, dependency state, Treaty state, outcomes, and evidence validity to recertification triggers.

Applicability: Operational Readiness, Verification, Certification, and Continuous Assurance. Responsible: Operations Assurance Lead. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

OPS-RDY-18-04 — Readiness approval

Readiness approval **MUST** be time- and scope-bounded and **MUST NOT** itself create operational or constitutional authority.

Applicability: Operational Readiness, Verification, Certification, and Continuous Assurance. Responsible: Operations Assurance Lead. Enforcement: procedure gate, policy check, monitoring, and independent evidence review. Evidence: signed operation record, correlated telemetry, decision log, verification result, and exception record where applicable. Severity: Critical. Exception authority: Operations Manager with Security and Assurance concurrence; higher-order requirements are not waivable. Verification: precondition test, negative-path exercise, runtime evidence, and independent closure review.

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Traceability and review

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Appendix A — Runbook and Companion Artifact Index

Appendix B — Certification Status

Book V v1.0 is certified final. Books VI, VIII, and IX are reconciled with no unresolved conflict or Owner Review item.